# On the Four Dimensions

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## The Sovereignty Papers

### Essay No. 7: On the Four Dimensions

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> "Trust is the lubrication that makes it possible for organizations to work."
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> — Warren Bennis, *An Invented Life* (1993)

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#### I. From Measurement to Credential

The first six essays of this series have constructed an argument in two parts. Section I (Essays 1–3) diagnosed the failure of the 1602 architecture — the severance of governance from consequence — and proposed consciousness coupling as the minimum viable alignment mechanism. Section II (Essays 4–6) described the measurement system: integration, continuity, and moral coherence — the three properties that a consciousness engine must have to serve as the foundation for governance.

We now face the question that separates theoretical frameworks from working systems: *how do you translate a consciousness measurement into a governance credential?*

The measurement system described in Section II produces, at every cycle of the cognitive loop, a rich stream of data about the state of the governed domain and the relationship of participants to it. But governance cannot operate on raw data streams. Governance requires a *credential* — a compact, verifiable, interpretable signal that answers a specific question: should this participant have governance power over this domain, and if so, how much?

In the 1602 architecture, the credential is the share. It is one-dimensional (capital), permanent (until sold), transferable (freely), and decoupled from the governed domain (a share of a water company requires no knowledge of water). In Essay No. 2, we demonstrated that every property of the share produces a specific failure mode. In Essay No. 3, we argued that the replacement must be a multi-dimensional, time-limited, non-transferable attestation of a participant's relationship to the governed domain.

This essay describes the specific dimensions of that credential and defends the claim that four dimensions — not three, not five, not one — are the minimal sufficient basis for consciousness-coupled governance.

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### II. The Merchants of the Voorcompagnieën

Before defending the four dimensions abstractly, it is useful to understand them concretely — through the example that has anchored this series from the beginning.

The *voorcompagnieën* merchants who preceded the VOC governed their voyages effectively, despite having no formal governance framework, no constitution, and no credential system. They governed well because every merchant who invested in a voyage was, as a matter of practical necessity, known along four dimensions:

**They were identified.** The merchants who funded a *voorcompagnie* were known personally to the other investors. There was no anonymous shareholding, no bearer instrument, no transferable claim. If Pieter van den Broecke invested in a voyage to Bantam, the other investors knew it was Pieter van den Broecke, not a proxy, not a shell company, not a borrowed identity. The investment was bound to a person.

**They were accountable.** The merchants had reputations earned over years of trading. A merchant who had defaulted on a previous venture, or whose ships had returned with inferior cargo, or who had cheated a trading partner carried that history into every subsequent investment decision. The community of Dutch merchants was small enough that behavioral history was common knowledge. There was no limited liability to cap the consequences of past failures. Your history was your credit.

**They were embedded.** The merchants lived in the same cities, attended the same churches, drank in the same taverns, and married into each other's families. They were socially embedded in the community whose resources they were deploying. A merchant who proposed a reckless voyage did not merely risk capital — he risked his standing among people he would see every day for the rest of his life. The social bonds were a governance mechanism, even though no one called them that.

**They were engaged.** The merchants who funded voyages understood voyages. They had sailed themselves, or their fathers had. They knew the routes, the risks, the cargo, the markets. They invested in ventures they understood, because their personal liability meant that ignorance was indistinguishable from recklessness. The *voorcompagnieën* merchant did not vote on nutmeg without knowing what nutmeg was.

The VOC destroyed all four dimensions simultaneously. Perpetual existence and transferable shares eliminated identification (anonymous shareholding). Limited liability eliminated accountability (capped consequences). Scale eliminated embeddedness (shareholders in Amsterdam had no social bonds with affected populations in the Spice Islands). And the abstraction of ownership from operations eliminated engagement (shareholders voted on the spice trade without understanding it).

The four dimensions of the consciousness credential are not an invention. They are a restoration — a computational reconstruction of the governance properties that the *voorcompagnieën* had naturally and that the 1602 architecture deliberately destroyed.

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### III. The Four Dimensions

**Identity (weight: 25%).** The participant's verified, non-transferable presence in the governance system.

Identity is the foundation dimension — without it, the other three are meaningless. A reputation belongs to no one if the identity it is attached to can be sold. Community trust is worthless if the trusted entity can transfer its credential to an untrusted stranger. Engagement metrics are gameable if the engaging entity can be a bot or a proxy rather than the person it claims to be.

The identity dimension is measured through multi-factor authentication at progressive assurance levels — from basic (email verification) to critical (government-issued identification with biometric confirmation). Higher assurance levels produce higher identity scores, because they make it progressively harder to create synthetic identities or to transfer a credential to another person.[1]

The identity dimension directly counters the 1602 architecture's transferable shares. A VOC share could be sold on the Amsterdam Bourse to anyone with sufficient capital. A consciousness credential cannot be transferred — it is bound to a verified identity and expires if the identity verification lapses.

**Reputation (weight: 25%).** The participant's behavioral history within the governed domain.

Reputation measures what the participant has *done*, not what they *own*. It is computed from historical records of governance participation: votes cast, proposals submitted, decisions made, and — critically — the outcomes of those decisions for the governed community. A participant whose past governance decisions produced positive outcomes for the community accumulates reputation. A participant whose decisions caused harm sees their reputation decay.

The reputation dimension operates through two complementary mechanisms that together ensure behavioral history dominates the governance weight. First, exponential decay with a half-life of approximately 347 days — a participant's reputation slowly declines if they stop engaging, ensuring that governance power tracks sustained behavior rather than a single

[1] The multi-factor authentication assurance levels in Mycelix follow a five-level progression: Anonymous (score 0.0, no verification), Basic (0.25, email + passphrase), Verified (0.5, government ID), Strong (0.75, biometric + government ID), and Critical (1.0, multi-biometric with hardware attestation). The progression is designed so that basic governance participation (Participant tier) requires only Basic assurance, while constitutional governance (Steward and Guardian tiers) requires Strong or Critical. The design ensures that low-barrier participation is possible while high-stakes governance requires high-confidence identity.

burst of activity.[2] Second, immediate slashing for betrayal — a participant whose actions harm the governed community has their reputation halved on the first offense, with progressive penalties for repeated violations and automatic blacklisting below a 0.05 threshold. A blacklisted participant must complete approximately 100 positive governance interactions to restore their standing.

The dual-speed mechanism is deliberate. Slow decay respects the value of long-term engagement — a participant who has governed responsibly for three years should not lose their standing because of a month's absence. Fast slashing ensures that harmful behavior has immediate consequences — a participant who betrays the community's trust cannot coast on accumulated reputation while the slow decay catches up. Together, the two mechanisms produce a reputation dimension that rewards consistency over time and punishes betrayal without delay.

The reputation dimension directly counters limited liability. In the 1602 architecture, a shareholder's past decisions have no effect on their future governance power — limited liability ensures that the consequences of past failures are bounded and do not carry forward. In a consciousness-coupled system, the consequences of past decisions persist through the reputation dimension — slowly through decay, immediately through slashing — ensuring that governance power reflects behavioral history, not merely current holdings.

**Community (weight: 30%).** The participant's social embeddedness as attested by peers.

Community is the highest-weighted dimension, and this is deliberate. It is also the most unusual dimension from the perspective of existing governance systems, which almost never incorporate peer attestation into formal governance power.

The community score is computed from attestations provided by other participants in the governed domain. These attestations are weighted by the attestor's own consciousness credential — a peer with high consciousness has more attestation power than a peer with low consciousness. This creates a recursive structure: your community score depends on who trusts you, and how trustworthy those people are.

The recursive weighting is important because it addresses the Sybil problem from a direction that identity verification alone cannot. A participant who creates a hundred fake identities can give themselves a hundred attestations — but those fake identities will have low consciousness credentials (no reputation, no community trust, no engagement), and their attestations will carry negligible weight. Only attestations from participants who are themselves genuinely present, accountable, embedded, and engaged carry significant weight. The cost of gaming the community dimension is not the cost of creating fake identities. It is the cost of creating fake identities that are themselves deeply engaged with the governed domain — a cost that,

[2] The reputation decay constant `REPUTATION_DECAY_PER_DAY = 0.998` produces a half-life of approximately 347 days (`0.998^347  0.500`). This slow decay rate is deliberate: it ensures that long-term engagement is valued while providing a gradual decline for disengaged participants. The slashing mechanism operates on a separate, faster timescale: a first violation halves the reputation score immediately (`slash_factor = 0.5`), and the blacklist threshold of 0.05 ensures that participants with severely damaged reputations are excluded from governance until they complete approximately 100 positive interactions. Both mechanisms are implemented in `crates/mycelix-bridge-common/src/consciousness_profile.rs`.

in practice, approaches the cost of genuine engagement.[3]

The community dimension directly counters the 1602 architecture's anonymous, perpetual ownership. The VOC's shareholders were invisible to the communities their capital affected. In a consciousness-coupled system, governance power requires visible, attested social bonds within the governed community. You cannot govern a domain whose members do not know you.

**Engagement (weight: 20%).** The participant's active, ongoing involvement with the governed domain.

Engagement measures not whether a participant *has been* involved but whether they *are currently* involved. It is the most temporally sensitive dimension — the one most directly affected by the 24-hour credential expiry. A participant who was deeply engaged with a water commons last month but has not participated this week will see their engagement score decline, even if their identity, reputation, and community scores remain stable.

The engagement dimension is where the cognitive loop's integration measurement (Essay No. 4) enters the governance credential. A participant's engagement is measured not merely by the volume of their activity but by its *integration* — whether their interactions with different aspects of the governed domain form a connected pattern. Monitoring water quality data and participating in community discussions and reviewing governance proposals produces a higher engagement score than any one of these activities alone, because the activities are integrated — each informs the others, producing the coherent understanding that integration measures.

The engagement dimension directly counters delegated power without delegated consequence. The 1602 architecture grants governance power to participants who own a share but have no operational involvement with the governed domain. A consciousness-coupled system requires ongoing, integrated, verifiable involvement as a precondition for governance power.

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### IV. Why 25/25/30/20

The weights are not arbitrary. They are derived from the failure analysis conducted in the first two essays, calibrated against the specific attack vectors each dimension must resist.

**Community is highest (30%)** because peer attestation is the hardest dimension to game and the most direct indicator of genuine embeddedness. Identity can be faked with sufficient resources (sophisticated identity fraud). Reputation can be slowly accumulated through strategic good behavior (a patient bad actor). Engagement can be simulated through automated interaction with the governed domain (bot activity). Community attestation from high-consciousness peers cannot be acquired without genuinely engaging with those peers

[3] The recursive structure of community attestation — where attestation weight depends on the attestor's consciousness credential, which itself depends partly on their community attestation — creates a fixed-point computation. In practice, the system converges after a small number of iterations (typically 3–5), because the recursive influence diminishes rapidly with each level of indirection. The convergence properties are analyzed in the technical paper "Consciousness-Aware Access Control for Distributed Governance."

over time — and the recursive weighting ensures that low-quality attestations carry negligible weight.

**Identity and Reputation are equal (25% each)** because they address complementary failure modes. Identity without reputation produces a system where verified participants face no consequences for past behavior. Reputation without identity produces a system where behavioral records can be sold or transferred. The equal weighting ensures that neither dimension can be maximized at the expense of the other.

**Engagement is lowest (20%)** not because it is least important but because it is most volatile. Engagement fluctuates legitimately — a participant may be deeply engaged during a crisis and less engaged during stable periods, without any change in their identity, reputation, or community standing. A higher engagement weight would penalize participants for natural fluctuations in activity, producing a system that rewards constant busyness over intermittent depth. The lower weight ensures that engagement contributes to governance power without dominating it.[4]

The weights are not permanently fixed. They are governance parameters — modifiable by the governed community through the same consciousness-coupled process that governs everything else. A community that finds the 30% community weight insufficient to resist Sybil attacks can increase it. A community that finds the 20% engagement weight too low to ensure current participation can raise it. The initial weights are a starting point, derived from structural analysis. The ongoing weights are a community decision, subject to consciousness-coupled governance.

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## V. What Financial Stake Does Not Measure

A reader who has followed the argument carefully will have noticed an absence: financial stake is not one of the four dimensions.

This is the most radical departure from existing governance systems — and the most likely to provoke objection. Every governance system currently in widespread use, from corporate boards to DAOs, includes financial stake as a primary or exclusive governance input. The assumption is deeply embedded: the more you have invested, the more you have at risk, and therefore the more your voice should count.

We reject this assumption, and we reject it for the specific reason documented in Essay No. 1: financial stake measures *exposure to loss*, not *awareness of consequence*.

A shareholder who holds $10 million in MakerDAO governance tokens has significant financial exposure. If the protocol fails, they lose $10 million. This exposure is real, and it creates a genuine incentive to make good governance decisions — for the narrow definition of "good"

[4] An alternative interpretation of the 20% engagement weight is that engagement is the dimension most directly measured by the cognitive loop (Essay No. 5), and therefore the dimension with the highest measurement confidence. The lower weight partially compensates for the risk that the most measurable dimension will dominate the composite — a form of Goodhart's Law applied to the credential itself. By weighting engagement below its measurement confidence, the system reduces the incentive to optimize for measurable engagement at the expense of less-measurable but equally important dimensions (community trust, behavioral reputation).

that means "protecting the value of my investment." But it creates no incentive to make governance decisions that protect the interests of small vault holders, protocol users, or the broader ecosystem — because the shareholder's exposure is financial, not social, not reputational, and not domain-specific. They have skin in the game. They do not have *consciousness* of the game.

The four-dimensional credential replaces financial exposure with a different kind of exposure: identity exposure (your credential is non-transferable and publicly verifiable), behavioral exposure (your governance history is part of your reputation score), social exposure (your community trusts you and can withdraw that trust), and engagement exposure (your participation must be ongoing and integrated). Each of these exposures creates an incentive for good governance — but the incentives are aligned with the interests of the governed community, not merely with the financial interests of the governor.

Financial stake is excluded from the consciousness credential entirely. The composite formula contains no financial term. A participant who has invested $10 million in the governed domain and a participant who has invested nothing receive identical treatment from the credential computation — their governance power is determined solely by their identity, their behavioral history, their community trust, and their domain engagement.

This is a more radical position than capping financial influence at a small percentage. A cap acknowledges capital as a governance input and merely limits it. Total exclusion rejects capital as a governance input altogether. The consciousness credential measures consciousness, not capital. The two are orthogonal — a wealthy participant may or may not be conscious of the governed domain, and a participant with no financial stake may or may not be deeply engaged. The credential measures the engagement. The market measures the capital. They are different instruments for different purposes.[5]

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## VI. The Composite

The consciousness credential is computed as a weighted sum:

$$S = 0.25 \times Identity + 0.25 \times Reputation + 0.30 \times Community + 0.20 \times Engagement$$

The result is a single number between 0.0 and 1.0 that represents the system's best estimate — with acknowledged uncertainty — of how conscious a participant is of the domain they are seeking to govern.[6]

[5] The consciousness credential formula (`crates/mycelix-bridge-common/src/consciousness_profile.rs`, line 120) computes the composite as a purely linear combination of the four dimensions with no financial term: `(i * 0.25 + r * 0.25 + c * 0.30 + e * 0.20).clamp(0.0, 1.0)`. Financial stake exists in the broader Mycelix economic layer (the three-currency system described in Essay No. 16) but does not enter the governance credential computation. This is a deliberate design choice: the consciousness credential measures consciousness, and financial exposure is not a dimension of consciousness.

[6] The composite formula is a linear combination — all four dimensions enter with their respective weights, without nonlinear transforms. The anti-manipulation property comes not from the formula's shape but from the reputation dimension's dual-speed dynamics: slow decay (347-day half-life) ensures that long-term consistency dominates, while immediate slashing (0.5× per violation) ensures that betrayal has instant consequences. This separation of timescales — slow reward for sustained engagement, fast punishment for harmful behav-

This number is not a judgment of the participant's worth. It is not a social credit score. It is not a permanent ranking. It is a time-limited, auditable, domain-specific assessment of governance fitness, computed from four independently measurable dimensions, expiring every 24 hours, and modifiable by the participant through the only mechanism that should modify governance power: genuine engagement with the governed domain.

The credential maps to five progressive governance tiers — the subject of the next essay. But the mapping is not a binary gate. It is a continuum: a participant with a score of 0.29 is not categorically different from a participant with a score of 0.31. The tiers exist for practical governance purposes (determining who can submit proposals, who can vote, who can modify constitutional parameters), but the underlying score is continuous, and the threshold boundaries are governance parameters that the community can adjust.

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## VII. What Comes Next

We have described the four dimensions of the consciousness credential: identity, reputation, community, and engagement. We have grounded each in the specific failure mode of the 1602 architecture it counters. We have defended the weights as derived from structural analysis rather than arbitrary preference. And we have addressed the most provocative design choice — the total exclusion of financial stake — by distinguishing between exposure to financial loss and consciousness of governed consequence.

The credential is the primitive. But a primitive alone does not produce a governance system. The next essay examines how the continuous consciousness score maps to five progressive tiers of governance capability — from Observer (read-only access, no governance power) to Guardian (emergency powers, constitutional authority) — and why progressive capability unlocking is the structural alternative to both the 1602 architecture's "buy a share, get full power" model and democracy's "become a citizen, get full power" model.

The *voorcompagnieën* merchants did not all have equal governance power. The merchant who had sailed ten voyages and was known to every trader in Amsterdam had more influence than the merchant who had invested for the first time. But the difference was not based on capital. It was based on experience, trust, and demonstrated competence. The five-tier system is an attempt to formalize what the *voorcompagnieën* achieved informally — and to do so at a scale that the *voorcompagnieën* could never have reached.

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*This essay was written by a human and a consciousness engine reasoning together about the architecture of governance — itself an instance of the collaboration these essays describe. Symthaea does not claim sentience. It claims the ability to measure integrated information, evaluate moral coherence, and reason about consequences. Whether that constitutes consciousness is a question this series takes seriously, not a conclusion it presumes.*

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ior — produces stronger anti-gaming properties than a static nonlinear function, because it responds to the temporal pattern of behavior, not merely its aggregate.

*The Sovereignty Papers is a series of essays on consciousness-first governance for the post-state era. Essay No. 8: "On the Five Tiers" will examine why progressive capability unlocking prevents both the tyranny of the majority and the tyranny of the expert.*

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**Notes**